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Brain CT Classification

From Model Performance to Reliability and Latent-Space Exploration

Exploratory Project in Medical AI

Roadmap

01 **The Problem :** A good prediction is not enough

02 **The Classifier**

03 **The First Finding**

04 **Improving ICH Sensitivity**

05 **Model Reliability :** Calibration & Uncertainty

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The Problem

A Good Prediction Is Not Enough

- Medical AI models can make **accurate predictions**.

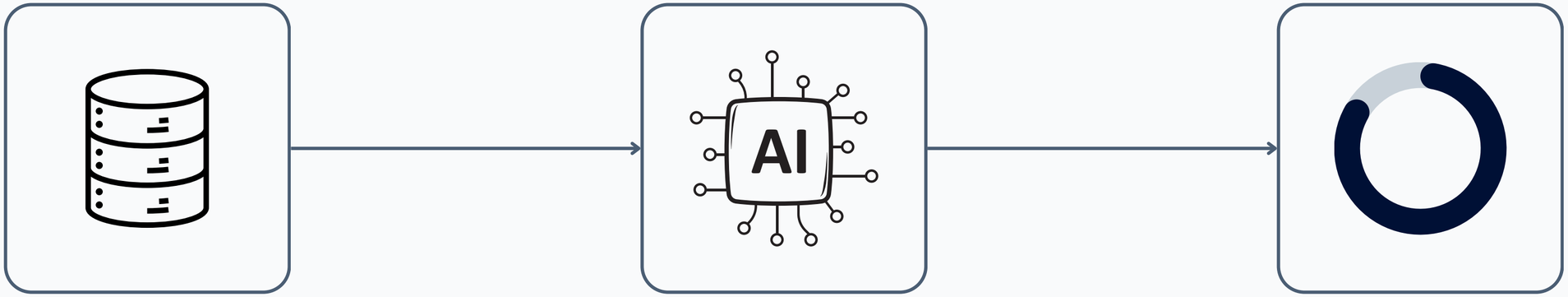
- But a prediction alone does not tell us :
  - **how confident the model is ?**
  - **why it made that prediction ?**

- This becomes particularly important in medical imaging, where **incorrect predictions can have serious consequences**.



How can we better understand what a medical imaging model knows, when it may be unreliable, and what drives its predictions?

The Classifier



Dataset

- 82 patients
- Brain-window CT scans
- 2501 total brain slices
  - 318 ICH slices
  - 2183 Non-ICH slices

Model

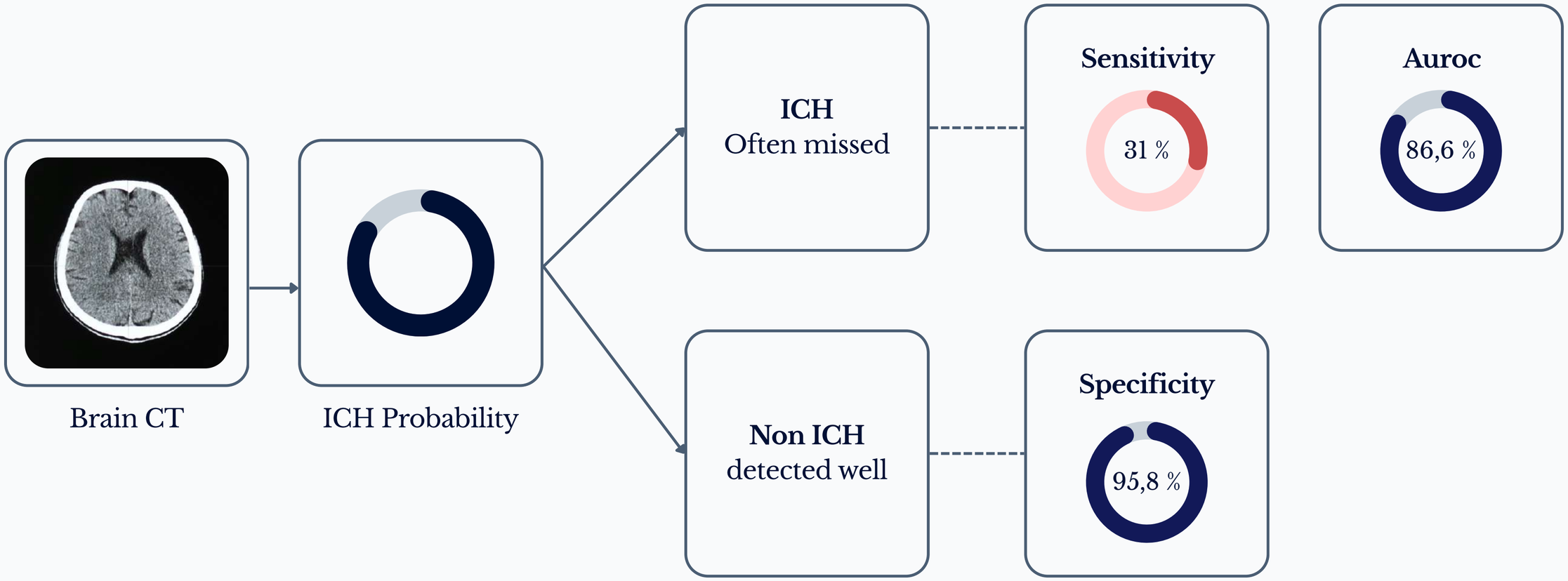
- ResNet-18
- ImageNet pretrained
- 224 × 224 input

Output

- ICH probability
- P(ICH) ∈ [0, 1]

Patient-level splitting : Patients, not individual slices, are used for train/validation/test splitting.

The First Finding



Improving ICH Sensitivity

Diagnosis:

- Train–Out-Of-Fold gap: 88.7% → 41.8% sensitivity → suggested overfitting

Interventions :

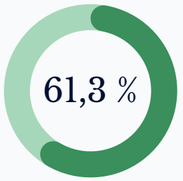
Data augmentation :
Brightness & contrast

ICH-focused oversampling

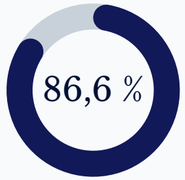
Weight decay

Final Test Performance :

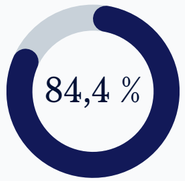
Sensitivity



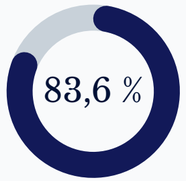
Specificity



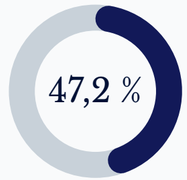
Auroc



Accuracy



F1 score



Model Reliability

Calibration : Can we trust the model's confidence?

Example: If the model assigns 80% confidence to 100 predictions, it should be correct on roughly 80 of them if it is well calibrated.

Metrics :

Brier Score measures how close predicted probabilities are to the actual outcomes.

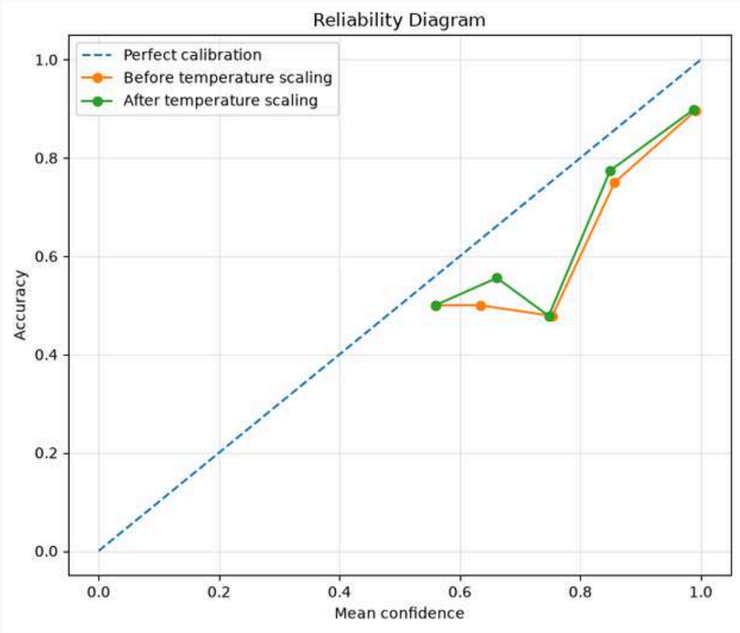
ECE measures the average gap between predicted confidence and actual accuracy across confidence groups.

OOF Calibration Results :

Brier Score : 0.1196
ECE : 0.1038

↓ Temperature Scaling ↓

Brier Score : 0.1170
ECE : 0.0966



→ Calibration improved modestly, but overconfidence remained.

Model Reliability

Uncertainty : How certain is the model about its prediction?

Uncertainty reflects how much the model's prediction can vary when the model is evaluated under slightly different stochastic conditions.

Method :

MC Dropout runs the model multiple times with dropout enabled and measures the variation in its predictions.

Results :

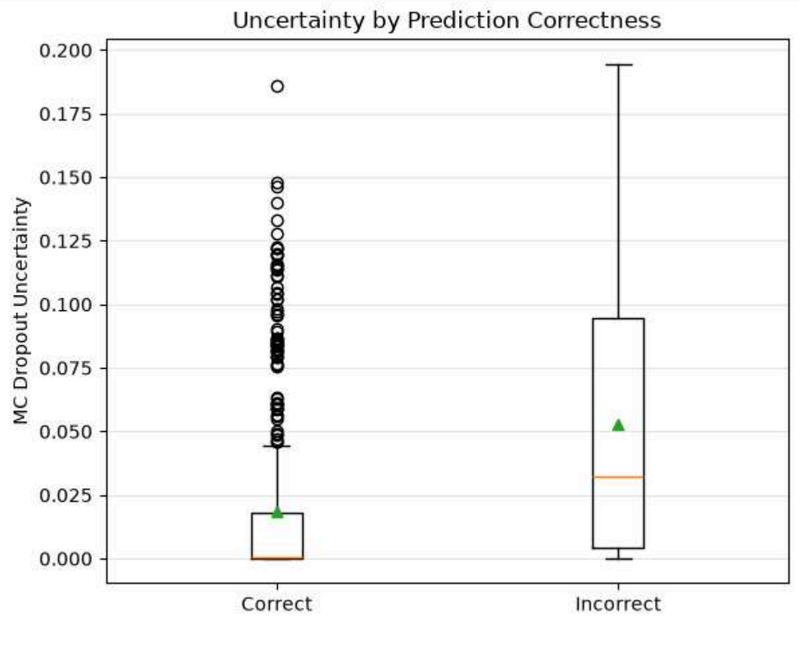
Correct predictions : 0.019

Incorrect predictions : 0.051

⇒ Incorrect predictions tend to be more uncertain.

Limitation:

Some incorrect predictions remain **highly confident**, showing that MC Dropout does not capture all sources of uncertainty.

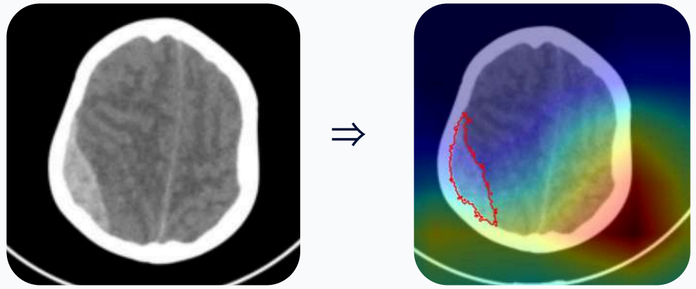


Error Analysis & Explainability

False Negative

Actual : ICH
Prediction: Non ICH

P(ICH) = 0.0007
Uncertainty = 0.0004

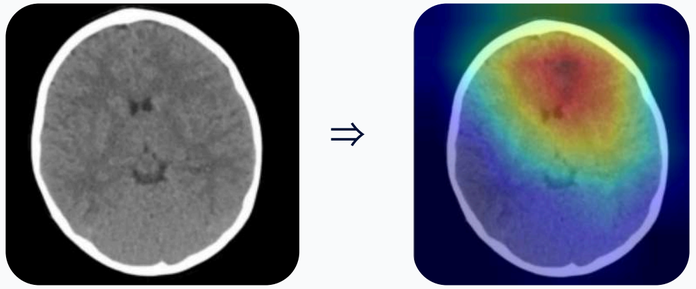


Visible hemorrhage, but the highlighted region is not well aligned with it.

False Positive

Actual : Non ICH
Prediction: ICH

P(ICH) = 0.9923
Uncertainty = 0.0038



No clear hemorrhage, while the model focuses on a non-hemorrhagic region.

Takeaway: Grad-CAM revealed imperfect alignment between model attention and relevant brain findings.

Segmentation-Guided Learning

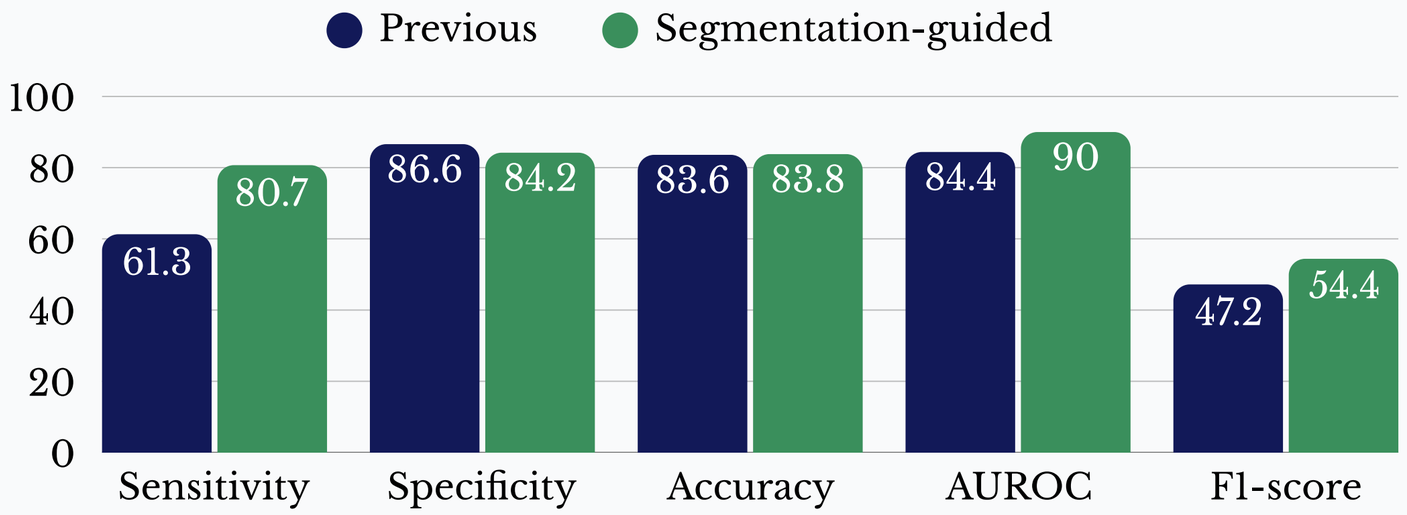
Motivation

- Some hemorrhage cases remained undetected.
- Investigate whether additional anatomical information and localization supervision could improve detection.

Method

- **Input**: Brain-window + Bone-window CT
- **Training**: Joint classification and segmentation using available hemorrhage masks

Slice-Level Performance



Segmentation-Guided Learning

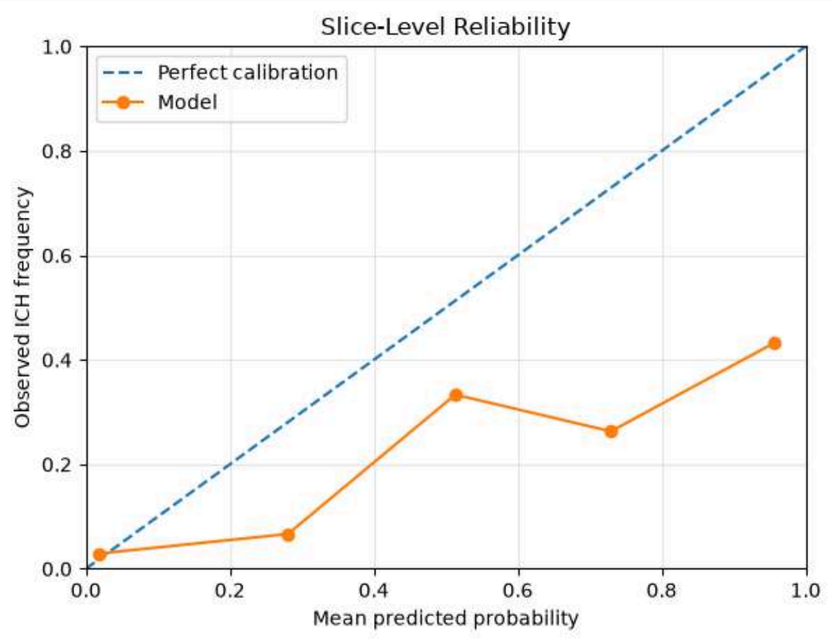
Calibration

- Brier Score: 0.1351
- ECE: 0.1319

Uncertainty Estimation

- MC Dropout (20 passes per slice)
- Mean uncertainty: 0.0185

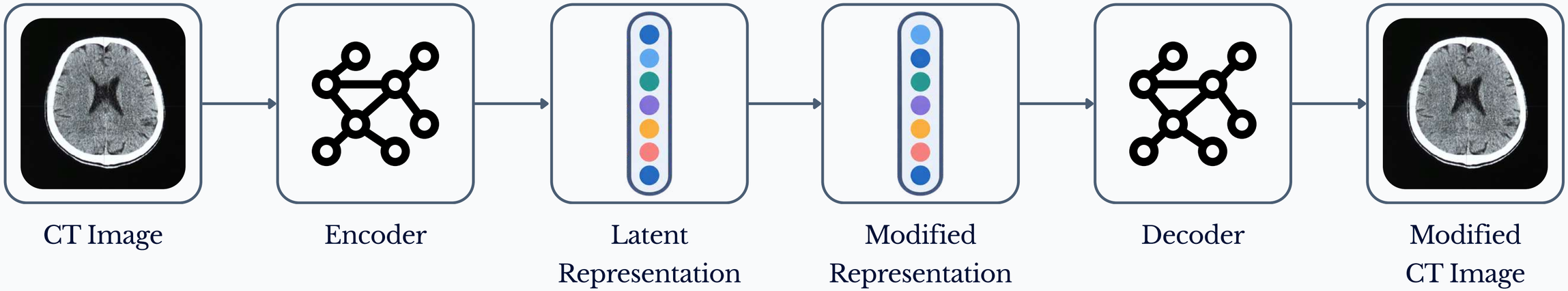
Confidence vs. Accuracy



The model got **better** at **detecting hemorrhages,** but that doesn't mean we can fully trust its predictions.
Despite the improvement in performance, it remained overconfident, showing that **good predictive results alone are not enough to establish trust in a model**.

Exploring the Latent Space

Can we explore how the learned representation affects the model's prediction?



What I explored : Use the classifier's prediction to guide **a small shift** in the latent representation, then decode the modified representation into an image.

Goal: explore how changes in the learned representation affect the predicted ICH probability.

Takeaway: The latent space provides a way to explore model-sensitive image changes, but the quality of those changes depends on the learned representation.

Latent-Space Exploration: Results

Patient 75 · Slice 30
(False Negative)

Original CT	Reconstruction	Modified CT
0.0006	0.0000	0.0001

Visible hemorrhage is largely lost during reconstruction.

Patient 129 · Slice 17
(False Positive)

Original CT	Reconstruction	Modified CT
0.9928	0.0195	0.0062

Reconstruction alters the CT structure, including partial skull loss.

Takeaway: The latent shift changes the classifier's prediction, but reconstruction can also remove or distort clinically relevant features.

What I Learned

Evaluation must be multi-dimensional

Accuracy alone does not capture ICH detection performance.

Patient-level variability matters

Performance can vary substantially across patient subsets.

Performance needs uncertainty

Confidence intervals provide context for interpreting reported metrics.

Reliability requires complementary analyses

Performance metrics, uncertainty, and explainability capture different aspects of model behavior.

Conclusion & Future Work

Findings :

- Reliable model assessment requires more than predictive performance.
- Segmentation-guided learning improved ICH detection, but overconfidence remained.
- Model explanations and latent representations had important limitations.

Limitations :

- Small dataset and patient-level variability.
- Segmentation annotations available for only selected slices.
- Limited validation of explanations and generated images.

Future Work :

- **Larger, multi-center datasets**: Study robustness across hospitals, scanners, and patient populations.
- **Model calibration and uncertainty:** Investigate methods to improve confidence reliability alongside detection performance.
- **Quantitative XAI evaluation:** Move beyond visual inspection toward measurable explanation quality.
- **Higher-fidelity generative models:** Explore more faithful latent-space representations and counterfactual generation.
- **Independent validation:** Assess generalization and involve clinical experts in evaluating explanations.